

RAYMOND (RAY) JOSEPH

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Staff ML/Research Engineer — AI Safety & Alignment — LLM Guardrails & Red Teaming — Founder with Exit
ML engineer with 12+ years building production AI safety systems, LLM guardrails, and evaluation frameworks across frontier AI, enterprise, regulatory, and classified government environments. Deep expertise in adversarial robustness, alignment techniques (RLHF, DPO, Constitutional AI), RAG architecture, red teaming, and deploying safety-critical ML at scale. Unique combination of hands-on safety engineering, published research, and founder experience shipping guardrail products to Fortune 500 clients.
Work authorization: US Work authorized; no sponsorship required, now or in the future.

TECHNICAL EXPERTISE

AI Safety & Alignment: Red teaming (jailbreak taxonomies, automated adversarial probing, attack surface mapping), guardrail systems (prompt injection defense, PII detection, toxicity filtering, factuality verification), alignment methods (RLHF, DPO, Constitutional AI, preference learning), bias auditing & fairness

LLM Systems & Evaluation: RAG pipelines (hybrid search, hierarchical chunking, retrieval optimization), LLM-as-judge frameworks, automated eval pipelines, hallucination detection, structured generation, safety benchmarking (TruthfulQA, HarmBench)

LLM Development: Fine-tuning (LoRA, QLoRA, RLHF), reward modeling, classifier-based content moderation, prompt engineering, confidence calibration, uncertainty quantification

ML Infrastructure: Ray, NVIDIA Triton, FastAPI, model serving at scale, real-time monitoring, A/B testing, drift detection, continuous retraining pipelines

Development Stack: Python, C++, SQL | PyTorch, TensorFlow, HuggingFace, LangChain | Personalization & recommendation systems

Specialized Models: LayoutLMv2, YOLOX, CLIP, Transformers | Multimodal systems, document & computer vision

WORK HISTORY

LegionIntel (legionintel.com)

Staff Software Engineer

12/2023 to Present, Austin, TX

- Architected production RAG systems processing **2M+ queries/month** with self-querying, hierarchical document chunking, and hybrid search, achieving **40% improvement** in answer relevance and citation accuracy. Deployed to **DoD IL4 (CUI) and IL6 (SECRET)** classified environments.
- Built comprehensive LLM evaluation and red teaming framework with automated adversarial testing pipelines, jailbreak detection, LLM-as-judge mechanisms, and structured generation for safety assurance. Reduced hallucinations by **94%** while maintaining user satisfaction above 4.5/5. All evaluation pipelines hardened to meet IL4/IL6 compliance and data handling requirements.
- Designed and deployed **multi-layer guardrail systems** for LLM applications in classified settings: prompt injection protection (rule-based + classifier ensemble), PII detection & redaction, toxicity filtering, and factuality verification with **<50ms latency overhead**. Guardrails architected to enforce strict data boundary controls required for IL6 SECRET-level operations.
- Developed systematic red teaming methodology including jailbreak taxonomy classification, automated adversarial prompt generation, and attack surface mapping. Used to validate safety properties and obtain **authorization to operate (ATO)** for AI systems in **DoD IL4 and IL6 environments**.
- Implemented feedback-based learning loops and continuous monitoring infrastructure that improved model accuracy by **31%** through systematic failure analysis, safety regression testing, and automated retraining pipelines. All within **air-gapped and FedRAMP-authorized infrastructure**.
- Led cross-functional initiatives with product, design, and engineering teams to deliver intelligent document analysis features from concept to production, securing **multi-million dollar DoD contracts** with live deployments across IL4 and IL6 classified environments.

Modguard.ai (See Forbes article)

Founder

01/2023 to 11/2023, Chicago, IL

- Founded AI safety startup building **production guardrail systems** for customer support chatbots and internal LLM tools, enabling safe, compliant conversational AI at scale for enterprise clients.
- Engineered modular guardrail framework with real-time content filtering, semantic safety checks, contextual policy enforcement, and **adversarial robustness testing** that integrated with existing customer service workflows via FastAPI microservices.
- Developed LoRA-finetuned safety classifiers and RLHF-aligned models achieving **97% harmful content detection** with **2% false positive rate**, applying Constitutional AI-inspired preference learning to balance helpfulness vs. safety for 3 enterprise deployments.
- Built secure RAG pipeline with source verification, privacy-preserving retrieval, and citation tracking; systematically evaluated against adversarial inputs (prompt injection, extraction attacks, topic hijacking); achieved product-market fit with Fortune 500 clients; **exited via acquisition**.

Financial Industry Regulatory Authority (FINRA)

Tech Lead and Manager (Corp Title: Product Lead/Manager)

9/2021 to 08/2023, Chicago, IL

- Led ML platform development ensuring **model safety, fairness, and regulatory compliance** across 3,400+ member firms, establishing governance frameworks with continuous validation, disparate impact testing, and drift monitoring.
- Architected bias detection and model auditing pipelines with human-in-the-loop review processes that improved model reliability by **31%**, creating explainability tools and fairness metrics that became **court-admissible evidence** for regulatory enforcement.
- Designed responsible AI oversight workflows including model cards, transparency reporting, and accountability mechanisms ensuring AI system integrity under regulatory and legal scrutiny. Collaborated cross-functionally with legal and engineering.

Barclays Capital

Quantitative Analytics (Vice President)

08/2015 to 08/2021, London, UK

- Designed ML systems for trading recommendations serving 400+ users with real-time anomaly detection, **confidence calibration**, and prediction monitoring across **50+ production models**, establishing uncertainty quantification methods.
- Built continuous monitoring and alerting pipelines for model degradation and data drift, establishing model risk management framework with **adversarial stress testing** and robustness validation that became bank-wide standard.
- Developed personalization and recommendation systems optimizing for user preferences while maintaining fairness constraints and regulatory compliance in high-stakes financial environments.

Additional Experience

- **Google Summer of Code, 2015:** Built robust numerical optimization algorithms with convergence guarantees; improved stability by **40%**.
- **Tilkee.fr, 2013:** Developed ML models with bias mitigation during internship.
- **Sauder Business School, 2012:** Researched statistical robustness and uncertainty quantification.

EDUCATION

IIT Madras

Aug 2010 to May 2015

Dual Degree: Bachelor's and Master's in Electrical Engineering

Madras, India

- Thesis estimated Delhi traffic congestion costs using AI simulations. Cited in **Indian Parliament and Uber campaigns**.

GPA: 8.33/10 (Cumulative); 8.05/10 (IITM only)

Institut National des Sciences Appliquées de Lyon

Sep 2013 to Dec 2014

Bachelor's Exchange in Big Data and Database Systems. GPA: 9.78/10

Lyon, France

Technische Universität München

May 2014 to July 2014

Master's Erasmus Exchange in Quantitative Analytics. GPA: 3.55/4

Munich, Germany

PUBLICATIONS & RESEARCH

- Published at **ACM/IEEE conferences** (ICSE, FSE) on software reliability, robustness, and system safety. Google Scholar.
- Research focus areas: **adversarial robustness**, uncertainty quantification, model reliability, and safety-critical system validation.
- Chaired sessions at **IEEE ICES, CICSyN, DMIN** on computational intelligence and system dependability.

PRESS COVERAGE

- **Modguard, AI Guardrails:** Forbes. Featured as a pioneering AI safety solution for enterprise LLM deployments.
- **GDP Forecasting using ML:** Forbes, Huffington Post
- **Delhi Traffic Study:** Economic Times, Business Insider